



DAT490 - Data
Science Capstone
Progress Report

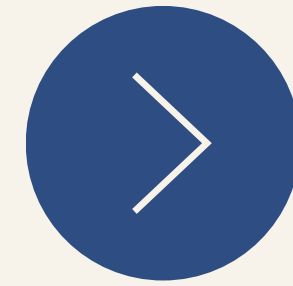


Analysis of Airline Cost Efficiencies

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Presentation Contents



01 Introduction

02 Literature
Review

03 Methodology

04 Current
Progress and
Next Steps



April 5th,
2026

Section

01

Introduction

April 5th,
2026

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Introduction

The U.S. airline industry is highly competitive and focused on controlling costs while maintaining efficiency and performance. Low-cost carriers are often seen as more efficient, while legacy airlines may provide better service, reliability, and more flight options.

We will discuss whether reducing costs actually improves operations or results in trade-offs that harm service quality. Does a lower operating cost lead to profit maximization in their domestic market?

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20–25%

The United States consistently dominates global air traffic, handling roughly 20–25% of worldwide air travel.





Motivating Problems

- Airlines try to reduce costs while maintaining reliable service
- Low-cost carriers are often seen as more efficient
- It is unclear whether lower costs improve or hurt performance

What is the relationship between airline cost efficiency and operational performance, and are there trade-offs between cost minimization and service quality?





Research Questions

1. How do different cost structures relate to operational efficiency measures such as load factor, capacity utilization, and passengers per flight?
2. Is there a relationship between operating costs and service performance metrics such as delays, reliability, and cancellations?
3. How do low-cost and legacy carriers differ in their balance between cost, efficiency, and performance?
4. How have these relationships changed pre-COVID, during COVID, and post-COVID periods?



Section

02

Literature Review

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Literature Review

The literature shows that airline efficiency is shaped by business model, network design, and cost structure. Low-cost carriers often appear more cost-efficient, but lower costs do not always translate to better service outcomes. Across the literature, a key open question is whether cost minimization improves operational performance or creates trade-offs in delays, cancellations, and reliability.

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1978 deregulation

U.S. airline deregulation increased competition and cost pressure



Rise of low-cost carriers

Low-cost and legacy carriers developed distinct business models



Modern competition and efficiency focus

Efficiency depends on utilization, network design, and scale



Controversies and Open Questions

Controversies

- Are delays a sign of inefficiency, or a byproduct of hub optimization?
- Are low-cost carriers always more efficient, or does this depend on how efficiency is measured?

Open questions

- Do lower operating costs improve performance or create service trade-offs?
- Did these relationships change after COVID?

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09



Section

03

Methodology.

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COVID-19 as a Structural Break

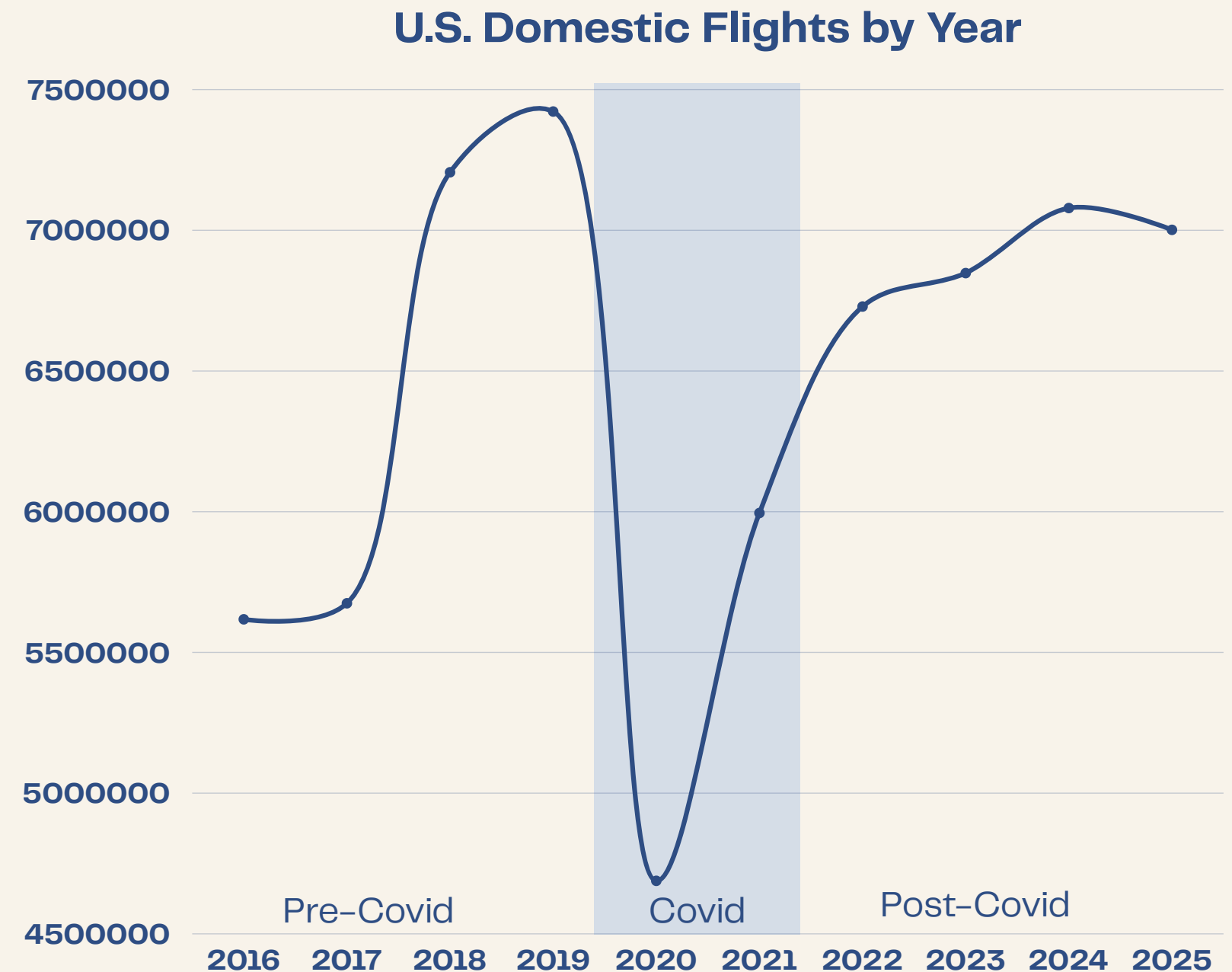
- Major demand shock
- Severe financial losses and disruptions
- Recovery patterns differed by airline type
- Justifies dividing analysis into:

Pre-Covid
2016-2019

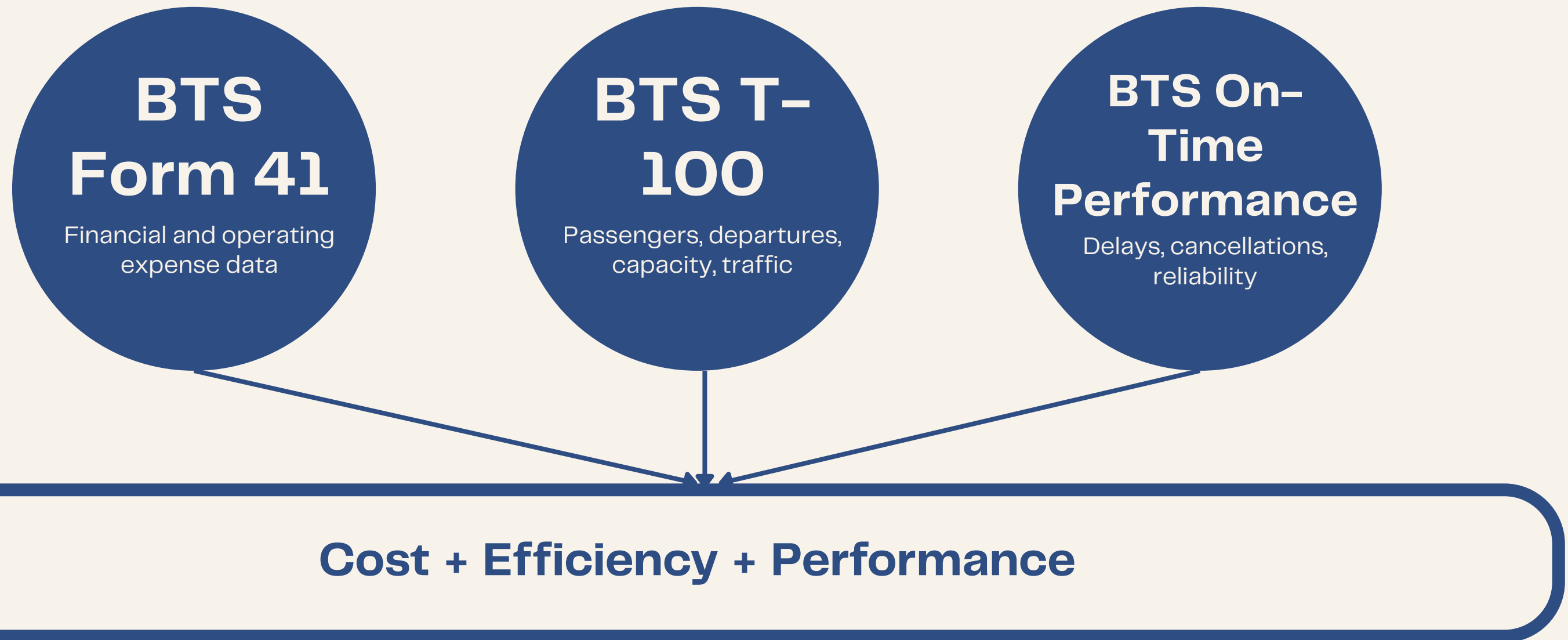
Covid
2020-2021

Post-Covid
2022-2025

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Major Datasets



Proposed Methodolog



Construct Indicators

- Cost per passenger
- Cost per departure
- Load factor
- Passengers per flight
- Delay rate
- Cancellation rate
- Reliability measures

Compare Airline Types

- Low-cost vs legacy carriers
- Pre-COVID vs COVID vs post-COVID

Analyze Relationships

- Regression models
- Comparative analysis
- Possible clustering of airlines by cost/performance profile



Examining Questions

<u>Key Question</u>	Finding a relationship between cost structures and operational efficiency metrics.	Observe the impact of operating costs on service performance indicators.	Predict the differences between the airline business models in balancing cost, efficiency, and performance.
<u>Metric</u>	CARRIER, OpExpenses, UNIQUE_CARRIER,	OpRevenues, TRAFFIC_EXPENSE, FOOD_EXPENSE	AIR_OP_EXPENSE NetIncome, OpProfitLoss,
<u>Definition</u>	Match the carrier ID with the route, aircraft, and segment. Obtain and coordinate data from datasets to develop into numeric variables.	Observe the indicator of Airline Profitability. Comparing cost per mile, cost per passenger, revenue per mile, and revenue per passenger.	Understand and measure route profitability. If the Airline is operating efficiently. Is the airline's pricing strategy profitable?



Section

04

Current Progress and Next Steps

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Current Progress + Next Steps

Completed

- Defined motivating question and sub-questions
- Reviewed literature on airline efficiency and performance
- Identified major datasets
- Developed preliminary methodology

Next Steps

- Clean and merge BTS datasets
- Construct KPIs
- Run initial descriptive analysis and regression models
- Refine comparisons by airline type and time period





Conclusion

Highlight
01

This project examines whether lower-cost airline strategies are associated with stronger operational outcomes or with trade-offs in service quality.

Highlight
02

The literature suggests that efficiency is shaped by business model, network design, and resource utilization, not by cost alone.

Highlight
03

Using BTS financial, traffic, and on-time performance data, the analysis compares airlines across pre-COVID, COVID, and post-COVID periods.

Highlight
04

The goal is to better understand how airlines balance cost, efficiency, and performance in a competitive and changing market.



Thank You!

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